8

ASSESSMENT METRICS FOR IMBALANCED LEARNING

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Abstract: Assessing learning systems is a very important aspect of the data-mining process. As such, many different types of metrics have been proposed over the years. In most cases, these metrics were not designed with the class imbalance problem in mind. As a result, some of them turn out to be appropriate for this problem, while others are not. The purpose of this chapter is to survey existing evaluation metrics and discuss their application to class-imbalanced domains. We survey many of the well-known metrics that were not specifically designed to handle class imbalances as well as more recent metrics that do take this issue into consideration.

8.1 INTRODUCTION

Evaluating learning algorithms is not a trivial issue as it requires judicious choices of assessment metrics, error-estimation methods, and statistical tests, as well as an understanding that the resulting evaluation can never be fully conclusive. This is due, in part, to the inherent bias of any evaluation tool and to the frequent violation of the assumptions they rely on [1]. In the case of class imbalances, the problem is even more acute because the default, relatively robust procedures used for unskewed data can break down miserably when the data is skewed.

Take, for example, the case of accuracy that measures the percentage of times a classifier predicts the correct outcome in a testing set. This simple measure is ineffective in the presence of class imbalances as demonstrated by the

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extreme case where, say, 99.9% of all the data is negative and only 0.1% positive. In such a case, the rough method, which consists of predicting "negative" in all cases, would produce an excellent accuracy rate of 99.9%. Obviously, this is not representative of what the classifier is really doing because, as suggested by its 0% recall, it is not an effective classifier at all, specifically if what it is trying to achieve is the recognition of rare, but potentially important events.

The inappropriateness of standard approaches is also apparent when considering error-estimation approaches. A 10-fold cross-validation, in particular, the most commonly used error-estimation method in machine learning, can easily break down in the case of class imbalances, even if the skew is less extreme than the one previously considered. For example, if 80% of the data is negative and 20% positive, when uniformly sampling the data in order to create the folds, it is quite likely that the distribution within each fold varies widely, with a possible dearth of positive data altogether in some extreme cases in which the dataset happens to be very small.

The behavior of statistical tests is altered as well by the presence of class imbalances, as the skew in the data can cause these tests to display different biases in different circumstances.

In this chapter, we will focus on the aspect of evaluation that concerns the choice of an assessment metric. This is, by far, the issue that has generated the greatest amount of attention in the community, as its answer is not as simple as the one pertaining to error estimation, where an easy solution consists of using stratified rather than purely random samplings of the data to create the folds (although there are other more complex issues with error estimation [1] and [2]); nor is it as complex as in the case of statistical testing, where the effect of class imbalances on a test can only be seen through numerous experiments and was not investigated at great length. We will, therefore, only touch upon the issues of error estimation and statistical testing briefly in the conclusion.

This chapter thus concentrates mainly on describing both metrics and graphical methods used in the case of class imbalances, concentrating on well-established methods and pointing out the newer experimental ones. Binary class problems will be the main focus, and the reader will be directed toward the literature that discusses extensions to multiple-class problems only in the conclusion.

The remainder of this chapter is organized as follows. In Section 8.2, we present an overview of the three families of assessment metrics used in machine learning—*threshold metrics, ranking methods and metrics* and *probabilistic metrics*—and discuss their general appropriateness to class imbalance situations. In Section 8.3, we focus on the threshold metrics particularly suited for imbalanced datasets. Section 8.4 looks at ranking methods and metrics often used in class-imbalanced situations. The probabilistic metrics did not seem specifically suited to the class imbalance problem; so we do not discuss them here. Finally, Section 8.5 concludes the chapter, looking at both other aspects of the classifier evaluation process that could be impacted on by class imbalances, and the case of multi-class-imbalanced problems.

8.2 A REVIEW OF EVALUATION METRIC FAMILIES AND THEIR APPLICABILITY TO THE CLASS IMBALANCE PROBLEM

Several machine learning researchers have identified three families of evaluation metrics used in the context of classification [3, 4]. These are the *threshold metrics* (e.g., accuracy and F-measure), the *ranking methods and metrics* [e.g., receiver operating characteristics (ROC) analysis and AUC), and the *probabilistic metrics* (e.g., root-mean-squared error). The purpose of this section is to discuss the advantages and disadvantages of these families with respect to the class imbalance problem.

In assessing both the families and the specific metrics, one must keep in mind the purpose of the evaluation process. For example, because Ferri et al. [3] considers that decreasing the overall value of an algorithm's worth because of its poor performance on one or a few infrequent classes is a nuisance, their conclusions are opposite to those typically reached by researchers who focus on problems with class imbalances or cost issues and for whom good performance on the majority class is usually of lesser consequence than good performance on the rarer or more important class. In this chapter, we are in agreement with the latter class of researchers, and so, in contrast to Ferri et al. [3], we take the position that sensitivity to the misclassification of infrequent classes is an asset rather than a liability.

As mentioned in [5], many studies have documented the weakness of the most notable threshold metric, accuracy, in comparison to ranking methods and metrics (most notably ROC analysis/AUC) in the case of class imbalances. The first such study, which in fact brought ROC analysis to the attention of machine learning researchers, is the one by Provost and Fawcett [6]. The main thrust of their article is that because the precise nature of the environment in which a machine learning system will be deployed is unknown, arbitrarily setting the conditions in which the system will be used is flawed. They argue that a typical measure such as accuracy does just that because it assumes equal error costs and constant and relatively balanced class priors, despite the fact that the actual conditions in which the system will operate are unknown. Their argument applies to more general cases than accuracy, but that is the metric they focus on in their discussion. Instead of systematically using the accuracy, they propose an assessment based on ROC analysis that does not make any assumptions about costs or priors, but rather, can evaluate the performance of different learning systems under all possible cost and prior situations. In fact, they propose a hybrid learning system based on the idea that suggests a different learner for each cost and prior condition.

A more recent study by Ferri et al. [3] expands this discussion to a large number of metrics and situations. It is the largest scale study, to date, which pits the different metric families and individual metrics against one another in various contexts (including the class imbalance one).[1] Their study is twofold. In

[1]A little earlier, Caruana and Niculescu-Mizil [4] also conducted a large-scale study of the different families of metrics, but they did not consider the case of class imbalances. We, thus, do not discuss their work here.

the first part, they determine the correlation between the different metrics and their family to other metrics and families, in various contexts; in the second part, they conduct a sensitivity analysis to study the specific performance of the group of metrics and their families that they studied in each context. We will focus only, herein, on the results they obtained in the context of the class imbalance problem.

The correlation study by Ferri et al. [3] shows that while the metrics are well correlated within each family in the balanced situation, this is not necessarily the case in the imbalanced situation. Indeed, two observations can be made: first, whatever correlations exist within a same family in the balanced case, these correlations are much weaker in the imbalanced situation, and second, the crossovers from one family to the next are different in the balanced and imbalanced cases. Without going into the details, these observations (and the first one in particular) allow us to conclude that the choice of metrics in the imbalanced case is of particular importance.

To analyze their sensitivity analysis results, we invert the conclusions of Ferri et al. [3], as previously mentioned, because we adopt the point of view that sensitivity to misclassification in infrequent classes is an asset rather than a liability. Note, however, that this inversion strategy should only be seen as a first approximation because the study by Ferri et al. [3] plots the probability of a wrong classification (as the class imbalance increases) no matter what class is considered and there is no way to differentiate between the different kinds of errors the classifiers make. Optimally, the study should be repeated from the perspective of researchers on the class imbalance problem. That being said, their study indicates that the rank-based measures behave best followed by some instances of threshold metrics. The probabilistic metrics that do behave acceptably well in the class imbalance case do so in the same spirit as some of the threshold metrics discussed later [the multi-class focused ones (Sections 8.3.5 and 8.3.6)]. In other words, it is not their probabilistic quality that makes them behave well in imbalanced cases, and therefore, they are not discussed any further in this chapter. The remainder of this chapter thus focuses only on the threshold metrics and ranking metrics most appropriate to the class imbalance problem. In the next two sections, we will discuss in great detail the measures that present specific interest for the class imbalance problem. We consider each class of metrics separately and discuss the particular metrics within these classes that are well suited to the class imbalance problem.

8.3 THRESHOLD METRICS: MULTIPLE- VERSUS SINGLE-CLASS FOCUS

As discussed in [1], threshold metrics can have a multiple- or a single-class focus. The multiple-class focus metrics consider the overall performance of the learning algorithm on all the classes in the dataset. These include accuracy, error rate, Cohen's kappa, and Fleiss' kappa measures. Precisely because of this multi-class focus, as also seen in the study by Ferri et al. [3], because the varying degree of importance on the different classes is not considered, performance metrics in this category do not fare very well in the class-imbalanced situation unless the class ratio is specifically taken into consideration. Single-class focus metrics, on the other hand, can be more sensitive to the issue of the varying degree of importance placed on the different classes and, as a result, be naturally better suited to evaluation in class-imbalanced domains. The single-class focus measures that are discussed in this section are: sensitivity/specificity, precision/recall, Geometric mean (G-mean), and F-measure. In addition to single-class focus metrics, we will discuss the multi-class focus metrics that take class ratios into consideration as a way to mitigate the contribution of the components on the overall results. We will also present a survey of more experimental metrics that were recently proposed but have not yet enjoyed much exposure in the community.

All the metrics discussed in this section are based on the concept of the confusion matrix. The confusion matrix for classifier f records the number of examples of each class that were correctly classified as belonging to that class by classifier f, as well as the number of examples of each class that were misclassified. For the misclassified examples, the confusion matrix considers all kinds of misclassification possible and records the number of examples that fall in each category. For example, if we consider a three-class problem, the following confusion matrix tells us that a examples of class A, e examples of class B, and i examples of class C were correctly classified by f. However, $b+c$ examples of class A were wrongly classified by f, b of which were mistakenly assigned to Class B, and c of which were mistakenly assigned to class C (and similarly for classes B and C).

	Predicted class A	Predicted class B	Predicted class C
Actual class A	a	b	c
Actual class B	d	e	f
Actual class C	g	h	i

In the binary class case, the above-mentioned matrix is reduced to a 2×2 format, and the issue of which class a misclassified example is assigned to disappears, as there remains only one possibility. In such a case, specific names are given to both the classes (positive and negative) and to the entries of the confusion matrix (true positive, false negative, false positive, and true negative) as shown in the following:

	Predicted positive	Predicted negative
Actual positive	True positive (TP)	False negative (FN)
Actual negative	False positive (FP)	True negative (TN)

The *true positive* and *true negative* entries indicate the number of examples correctly classified by classifier f as positive and negative, respectively. The *false negative* entry indicates the number of positive examples wrongly classified as negative. Conversely, the *false positive* entry indicates the number of negative examples wrongly classified as positive. With these quantities defined, we are now able to define the single-class focus metrics of interest here as well as the multiple-class focus metrics adapted to the class imbalance problem.

8.3.1 Sensitivity and Specificity

The *sensitivity* of a classifier f corresponds to its true positive rate or, in other words, the proportion of positive examples actually assigned as positive by f. The complement metric to this is called the *specificity* of classifier f and corresponds to the proportion of negative examples that are detected. It is the same quantity, only it is measured over the negative class. These two metrics are typically used in the medical context to assess the effectiveness of a clinical test in detecting a disease. They are defined as follows:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8.1}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}} \tag{8.2}$$

These measures, together, identify the proportions of the two classes correctly classified. However, unlike accuracy, they do this separately in the context of each individual class of instances. As a result, the class imbalance does not affect these measures. On the other hand, the cost of using these metrics appears in the form of a metric for each single class, which is more difficult to process than a single measure. Another issue that is missed by this pair of metrics is the measure of the proportion of examples assigned to a given class by classifier f that actually belongs to this class. This aspect is captured by the following pair of metrics.

8.3.2 Precision and Recall

The *precision* of a classifier f measures how precise f is when identifying the examples of a given class. More precisely, it assesses the proportion of examples assigned a positive classification that are truly positive. This quantity together with sensitivity, which is commonly called *recall* when considered together with precision, is typically used in the information retrieval context where researchers are interested in the proportion of relevant information identified along with the amount of actually relevant information from the information assessed as relevant by f. Precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8.3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8.4}$$

Here again, the focus is on the positive class only, meaning that the problems encountered by multi-class focus metrics in the case of the class imbalance problem are, once more, avoided. As for sensitivity and specificity, however, the cost of using precision and recall is that two measures must be considered and that absolutely no information is given on the performance of f on the negative class. This information did appear in the form of specificity in the previous pair of metrics.

We now turn our attention to a couple of ways of combining the components of the two pairs of single-focus metrics just discussed so as to obtain a single measure instead of a pair of metrics.

8.3.3 Geometric Mean

The G-mean was introduced by Kubat et al. [7] specifically as a response to the class imbalance problem and as a response to the fact that a single metric is easier to manipulate than a pair of metrics. This measure takes into account the relative balance of the classifier's performance on both the positive and the negative classes. In order to do so, it is defined as a function of both the sensitivity and the specificity of the classifier. G-mean is defined in more detail as follows:

$$G - \text{mean}_1 = \sqrt{\text{sensitivity} \times \text{specificity}} \tag{8.5}$$

Because the two classes are given equal importance in this formulation, the G-mean, while more sensitive to class imbalances than accuracy, remains close, in some sense, to the multi-class focus category of metrics. Another version of the G-mean was, therefore, also suggested, which focuses solely on the positive class. In order to do so, it replaces the specificity term by the precision term, yielding

$$G - \text{mean}_2 = \sqrt{\text{sensitivity} \times \text{precision}} \tag{8.6}$$

8.3.4 F-Measure

The F-measure is another combination metric whose purpose this time is to combine the values of the precision and recall of a classifier f on a given domain to a single scalar. It does so in a more sophisticated way than the G-mean, as it allows the user to weigh the contribution of each component as desired. More specifically, the F-measure is the weighted harmonic mean of precision and recall, and is formally defined as follows: For any $\alpha \in R$, $\alpha > 0$,

$$F_\alpha = \frac{(1+\alpha)[\text{precision} \times \text{recall}]}{[\alpha \times \text{precision}] + \text{recall}} \tag{8.7}$$

where α typically takes the values of 1, 2, or 0.5 to signify that precision and recall are equal in the case of $\alpha = 1$, that recall weighs twice as much as precision when $\alpha = 2$ and that precision weighs twice as much as recall when $\alpha = 0.5$. The F-measure is very popular in the domains of information retrieval or text categorization.

We now get back to multi-class focus metrics but show how their shortcomings in the case of class imbalances can be overcome with appropriate weightings.

8.3.5 Macro-Averaged Accuracy

The macro-averaged accuracy (MAA) is presented in [3] and is calculated as the *arithmetic* average of the partial accuracies of each class. Its formula is given as follows for the binary case ([3] gives a more general definition for multi-class problems):

$$\text{MAA} = \frac{\text{sensitivity} + \text{specificity}}{2} \tag{8.8}$$

In this formulation, we see that no matter what the frequency of each class is, the classifier's accuracy on each class is equally weighted.

The second macro-averaged accuracy presented in [3] is the macro-averaged accuracy calculated as the *geometric* average of the partial accuracies of each class and is nothing but the G-mean that was already presented in Section 8.3.3. Both measures were shown to perform better than other threshold metrics in class-imbalanced cases in the experiments from [3] that we related in Section 8.2.

8.3.6 Newer Combinations of Threshold Metrics

Up to this point, we have only discussed the most common threshold metrics that deal with the class imbalance problem. In the past few years, a number of newer combinations of threshold metrics have been suggested to improve on the ones previously used in the community. We discuss them now.

8.3.6.1 Mean-Class-Weighted Accuracy The mean-class-weighted accuracy [8] implements a small modification to the macro-average accuracy metric just presented. In particular, rather than giving sensitivity and specificity equal weights, Cohen et al. [8] choose to give the user control over each component's weight. More specifically, Cohen et al. [8] who applied machine learning methods to medical problems found that both the G-mean and the F-measure were insufficient for their purposes. On the one hand, they found that the G-mean does not provide a mechanism for giving higher priority to rare instances coming from the minority class; on the other hand, they found that while the F-measure allows for some kind of prioritization, it does not take into account the performance on the negative class. In medical diagnosis tasks, Cohen et al. [8] claim that what would be useful is a relative weighting of sensitivity and specificity. They are able to achieve this by using the macro-average, but adding a user-defined weight to increase the emphasis given to one class over the other. The measure they propose is called the *mean-class-weighted accuracy* (MCWA) and is defined as follows for the binary case:

$$\text{MCWA} = w \times \text{sensitivity} + (1 - w) \times \text{specificity} \tag{8.9}$$

where w is a value between 0 and 1, which represents the weight assigned to the positive class. As for the macro-averages described in Section 8.3.5, the formula is also defined for multiple classes. This newer measure, they found, fits their purpose better than the other existing combinations.

In the next section, we present yet another variation on the sensitivity/specificity combination.

8.3.6.2 Optimized Precision Optimized precision was proposed by Ranawana and Palade [9] as another combination of sensitivity and specificity. This time, the goal is not to weigh the contributions of the classes according to some domain requirement, but rather to optimize both sensitivity and specificity. The new metric is defined as follows:

$$\begin{aligned}\text{Optimized precision} = \text{specificity} \times N_{\text{n}} &+ \text{sensitivity} \\ &\times N_{\text{p}} - \frac{|\text{specificity} - \text{sensitivity}|}{\text{specificity} + \text{sensitivity}}\end{aligned} \tag{8.10}$$

where N_n represents the number of negative examples in the dataset, and N_p represents the number of positive examples in the dataset. The authors show that this measure is preferable to precision and other single performance measures, both theoretically and in an experimental study, but they do not compare it to the other combination metrics that were already discussed in the previous sections.

8.3.6.3 Adjusted Geometric Mean The adjusted geometric mean (AGm) was designed to deal with the type of class imbalance problems whose purpose is to increase the sensitivity while keeping the reduction of specificity to a minimum. This is in contrast with the problems about which we do not have to worry as much about a reduction in specificity. Batuwita and Palade [10], the designers of AGm, showed that commonly used measures in imbalanced datasets are not adequate for their problem of interest in the domain of bioinformatics. The G-mean may lead the researcher to choose a model that reduces specificity too much; as for the F-measure, although it could, theoretically, avoid this problem by manipulating its parameter, it is not clear what value that parameter should take when the misclassification costs are not exactly known. To overcome these problems, Batuwita and Palade [10] proposed the AGm. Its definition is as follows:

$$\mathrm{AGm} = \frac{\mathrm{G-mean} + \mathrm{specificity} \times N_{\mathrm{n}}}{1 + N_{\mathrm{n}}}, \text{ ifspecificity} > 0 \tag{8.11}$$

$$\mathrm{AGm} = 0, \text{ ifsensitivity} = 0 \tag{8.12}$$

This metric is more sensitive to variations in specificity than in sensitivity. Furthermore, this focus on specificity is related to the number of negative examples in the dataset so that the higher the degree of imbalance, the higher the reaction to changes in specificity. The authors showed the advantages of this new measure as compared to the G-mean and the F-measure.

8.3.6.4 Index of Balanced Accuracy The designers of the index of balanced accuracy (IBA) [11] respond to the same type of problem of the G-mean (as well as of the AUC) described by Batuwita and Palade [10]. In particular, they show that different combinations of sensitivity and specificity can lead to the same values for the G-mean. Their generalized IBA is defined as follows:

$$\mathrm{IBA}_{\alpha}(M) = (1 + \alpha \times \mathrm{Dom}) \times M \tag{8.13}$$

where dominance (Dom) is defined as:

$$\mathrm{sensitivity} - \mathrm{specificity} \tag{8.14}$$

where M is any metric, and α is a weighting factor designed to reduce the influence of the dominance on the result of a particular metric M. IBA_{α} was tested both experimentally on artificial and University of California, Irvine (UCI) data and theoretically. It is shown to have low correlation with accuracy (which is not appropriate for class-imbalanced datasets), but high correlation with G-mean and AUC (which are appropriate for class-imbalanced datasets). It is, therefore, indeed, geared at the right type of problems.

8.4 RANKING METHODS AND METRICS: TAKING UNCERTAINTY INTO CONSIDERATION

An important disadvantage of all the threshold metrics discussed in the previous section is that they assume full knowledge of the conditions under which the classifier will be deployed. In particular, they assume that the class imbalance present in the training set is the one that will be encountered throughout the operating life of the classifier. If that is truly the case, then the previously discussed metrics are appropriate; however, it has been suggested that information related to skew (as well as cost and other prior probabilities) of the data is generally not known. In such cases, it is more useful to use evaluation methods that enable visualization or summarization performance over the full operating range of the classifier. In particular, such methods allow for the assessment of a classifier's performance over all possible imbalance or cost ratios.

In this section, we will discuss ROC analysis, cost curves, precision–recall (PR) curves, as well as a summary scalar metric known as *AUC or AUROC* (Area under the ROC curve). In addition, newer and more experimental methods and metrics will be presented.

8.4.1 ROC Analysis

In the context of the class imbalance problem, the concept of ROC analysis can be interpreted as follows. Imagine that instead of training a classifier f only at a given class imbalance level, that classifier is trained at all possible imbalance levels. For each of these levels, two measurements are taken as a pair, the true positive rate (or sensitivity), whose definition was given in Equation 8.1, and the false positive rate (FPR) (or false alarm rate), whose definition is given by:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{8.15}$$

Many situations may yield the same measurement pairs, but that does not matter as duplicates are ignored. Once all the measurements have been made, the points represented by all the obtained pairs are plotted in what is called the *ROC space*, a graph that plots the true positive rate as a function of the false positive rate. The points are then joined in a smooth curve, which represents the ROC curve for that classifier. Figure 8.1 shows two ROC curves representing the performance of two classifiers *f1* and *f2* across all possible operating ranges.[2]

The closer a curve representing a classifier f is from the top left corner of the ROC space (small false positive rate, large true positive rate) the better the performance of that classifier. For example, $f1$ performs better than $f2$ in the graph of Figure 8.1. However, the ideal situation of Figure 8.1 rarely occurs in practice. More often than not, one is faced with a situation as that of Figure 8.2, where one classifier dominates the other in some parts of the ROC space, but not in others.

The reason why ROC analysis is well suited to the study of class imbalance domains is twofold. First, as in the case of the single-class focus metrics of the previous section, rather than being combined together into a single multi-class focus metric, performance on each class is decomposed into two distinct measures. Second, as discussed at the beginning of the section, the imbalance

[2]Note that this description of ROC analysis is more conceptual than practical. The actual construction of ROC curves uses a single training set and modifies the classifier's threshold to generate all the points it uses to build the ROC curve. More details can be found in [1].

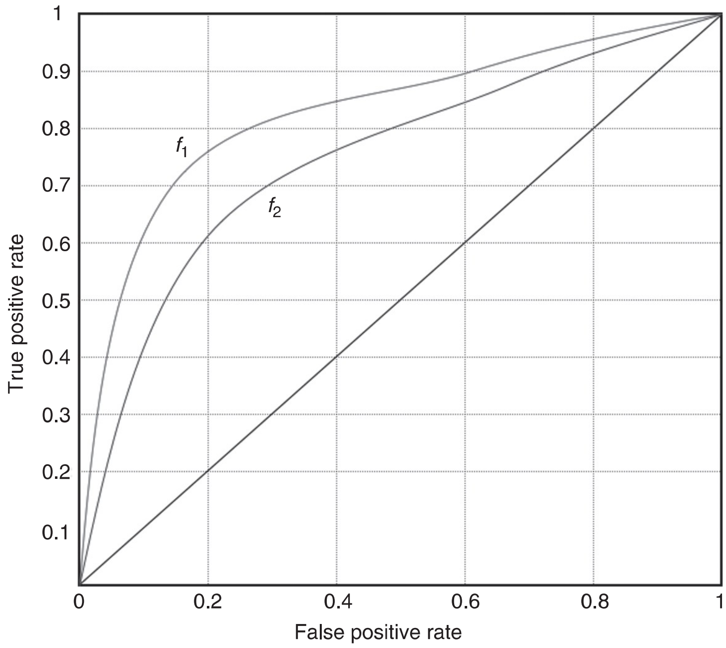

Figure 8.1 The ROC curves for two hypothetical scoring classifiers f_1 and f_2.

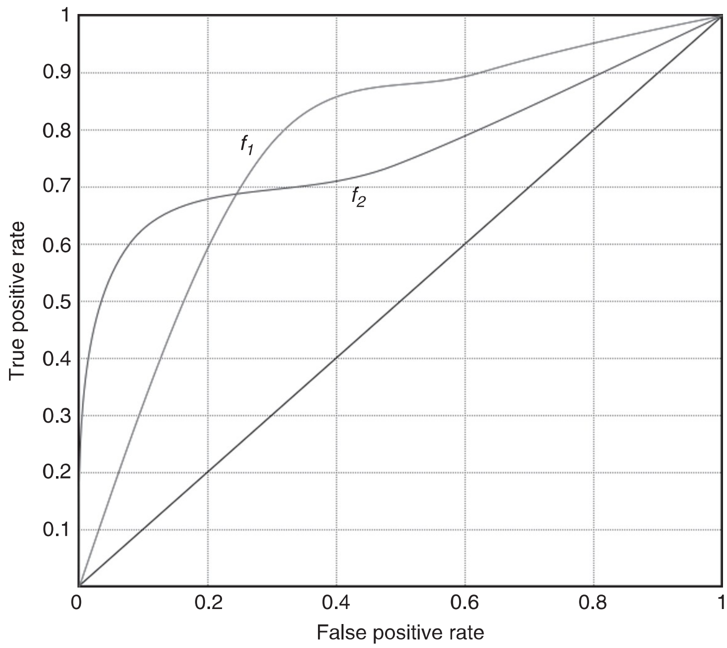

Figure 8.2 The ROC curves for two hypothetical scoring classifiers f_1 and f_2, in which a single classifier is not strictly dominant throughout the operating range.

ratio that truly applies in a domain is rarely precisely known. ROC analysis gives an evaluation of what may happen in these diverse situations.

ROC analysis was criticized by Webb and Ting [12] because of the problems that may arise in the case of changing distributions. The issue is that the true and false positive rates should remain invariant to changes in class distributions in order for ROC analysis to be valid. Yet, Webb and Ting [12] argue that changes in class distributions often also lead to changes in the true and false positive rates, thus invalidating the results. A response to Webb and Ting [12] was written by Fawcett and Flach [13], which alleviates their worries. Fawcett and Flach [13] argue that within the two important existing classes of domains, the problems pointed out by Webb and Ting [12] only apply to one of these classes, and then not always. When dealing with class imbalances in the case of changing distributions, it is thus recommended to be aware of the issues discussed in these two papers, before fully trusting the results of ROC analysis. In fact, Landgrebe et al. [14] point out another aspect of this discussion. They suggest that in imprecise environments, when class distributions change, the purpose of evaluation is to observe variability in performance, as the distribution changes in order to select the best model within an expected range of priors. They argue that ROC analysis is not appropriate for this exercise, and, instead, suggest the use of PR curves. PR curves are discussed in Section 8.4.3, which is followed in Section 8.4.5.2 by Landgrebe et al. [14]'s definition of PR curves summaries, which is somewhat similar to that of the AUC.

In the meantime, the next section addresses another common complaint about ROC analyses, namely that they are not practical to read in the case where a class imbalance or cost ratio is known.

8.4.2 Cost Curves

The issue of ROC analysis readability is taken into account by cost curves. What makes cost curves attractive is their ease of use in determining the best classifier to use in situations where the error costs or class distribution, or more generally the skew, are known. For example, in Figure 8.2, while it is clear that the curve corresponding to classifier f_2 dominates the curve corresponding to classifier f_1 at first and that the situation is inverted afterward, this information cannot easily be translated into information telling us for what costs and class distributions classifier f_2 performs better than classifier f_1. Cost curves do provide this kind of information.

In particular, cost curves plot the relative expected misclassification costs as a function of the proportion of positive instances in the dataset. An illustration of a cost curve plot is given in Figure 8.3. The important thing to keep in mind is that there is a point-line duality between cost curves and ROC curves. Cost curves are point-line duals of the ROC curves. In ROC space, a discrete classifier is represented by a point. The points representing several classifiers (produced by manipulating the threshold of the base classifier) can be joined (and extrapolated) to produce a ROC curve. In cost space, each of the ROC points is represented by

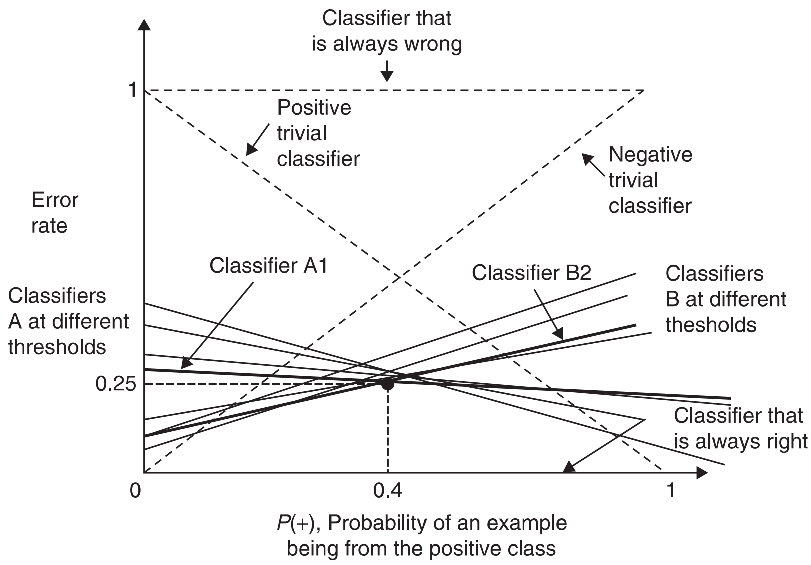

Figure 8.3 The cost curves illustration for two hypothetical scoring classifiers A_1 and B_2, in which a single classifier is not strictly dominant throughout the operating range.

a line, and the convex hull of the ROC space corresponds to the lower envelope created by all the classifier lines.

In cost space, the relative expected misclassification cost is plotted as a function of the probability of an example belonging to the positive class. Since the point in the ROC space where two curves cross is also represented in cost space by the crossing of two lines, this point can easily be found in cost space and its x-coordinate read off the graph. This value corresponds to the imbalance ratio at which a switch from the classifier that dominated the ROC space up to the curve crossing (say, classifier A) to the classifier that started dominating the ROC space afterward (say, classifier B) is warranted. The cost curves thus make it very easy to obtain this information. In contrast, ROC graphs tell us that, sometimes, A is preferable to B, but one cannot read off when this is so, from the ROC graph.

8.4.3 Precision–Recall Curves

PR curves are similar to ROC curves in that they explore the trade-off between the well-classified positive examples and the number of misclassified negative examples. As the name suggests, PR curves plot the precision of the classifier as a function of its recall, as shown in Figure 8.4. In other words, it measures the amount of precision that can be obtained as various degrees of recall are considered. For instance, in the domain of document retrieval systems, PR curves would plot the percentage of relevant documents identified as relevant against the percentage of relevant documents deemed as such with respect to all the

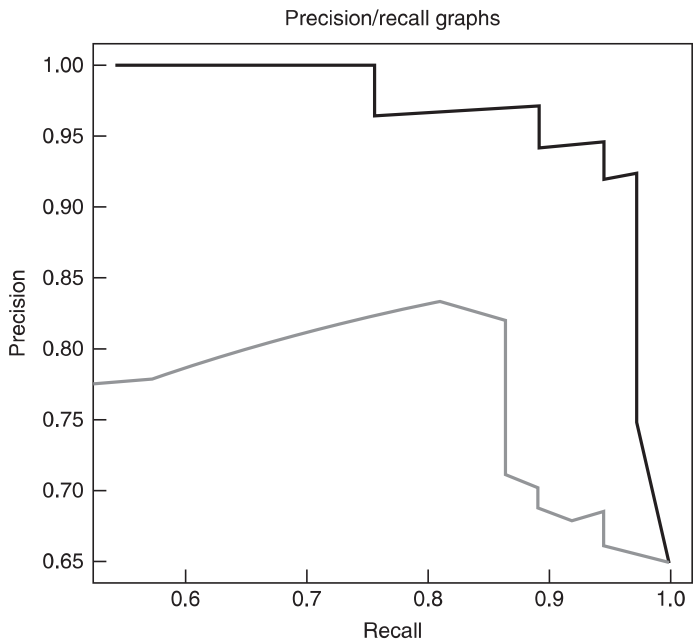

Figure 8.4 The precision–recall curves illustration for two scoring classifiers.

documents in the sample. The curves, thus, look different from ROC curves as they have a negative slope. This is because precision decreases as recall increases. PR curves are a popular visualization technique in the information retrieval field as illustrated by our earlier examples that discussed the notions of precision and recall. Further, it has been suggested that PR curves are sometimes more appropriate than ROC curves in the event of highly imbalanced data [15].

8.4.4 AUC

The AUC represents the performance of a classifier averaged over all the possible cost ratios. Noting that the ROC space is a unit square, it can be clearly seen that the AUC for a classifier f is such that $\mathrm{AUC}(f) \in [0, 1]$ with the upper bound attained for a perfect classifier (one with TPR $= 1$ and FPR $= 0$). Moreover, as can be noted, the random classifier represented by the diagonal cuts the ROC space in half and hence $\mathrm{AUC}(f_{\mathrm{random}}) = 0.5$. For a classifier with a reasonably better performance than random guessing, we would expect to have an AUC greater than 0.5.

Elaborate methods have been suggested to calculate the AUC. However, using the Wilcoxon's rank sum statistic, we can obtain a simpler manner of estimating the AUC for ranking classifiers. To the scores assigned by the classifier to each test instance, we associate a rank in the order of decreasing scores. That is, the example with the highest score is assigned the rank 1. Then, we can calculate the AUC as:

$$\mathrm{AUC}(f) = \frac{\sum_{i=1}^{|T_\mathrm{p}|}(R_i - i)}{|T_\mathrm{p}||T_\mathrm{n}|}$$

where $T_\mathrm{p} \subset T$ and $T_\mathrm{n} \subset T$ are, respectively, the subsets of positive and negative examples in test set T, and R_i is the rank of the ith example in T_p given by classifier f.

AUC basically measures the probability of the classifier assigning a higher rank to a randomly chosen positive example than a randomly chosen negative example. Even though the AUC attempts to be a summary statistic, just as other single metric performance measures, it too loses significant information about the behavior of the learning algorithm over the entire operating range (for instance, it misses information on concavities in the performance, or trade-off behaviors between the TP and FP performances).

It can be argued that the AUC is a good way to get a score for the general performance of a classifier and to compare it to that of another classifier. This is particularly true in the case of imbalanced data where, as discussed earlier, accuracy is too strongly biased toward the dominant class. However, some criticisms have also appeared warning against the use of AUC across classifiers for comparative purposes. One of the most obvious ones is that if the ROC curves of the two classifiers intersect (such as in the case of Figure 8.2), then the AUC-based comparison between the classifiers can be relatively uninformative and even misleading. However, a possibly more serious limitation of the AUC for comparative purposes lies in the fact that the misclassification cost distributions (and hence the skew-ratio distributions) used by the AUC are different for different classifiers. This is discussed in the next subsection, which generally looks at newer and more experimental ranking metrics and graphical methods.

8.4.5 Newer Ranking Metrics and Graphical Methods

8.4.5.1 The H-measure The more serious criticism of AUC just mentioned means that, when comparing different classifiers using the AUC, one may in fact be comparing oranges and apples, as the AUC may give more weight to misclassifying a point by classifier A than it does by classifier B. This is because the AUC uses an implicit weight function that varies from classifier to classifier. This criticism was made by Hand [16], who also proposed the H-measure to remedy this problem. The H-measure allows the user to select a cost-weight function that is equal for all the classifiers under comparison and thus allows for fairer comparisons. The formulation of the H-measure is a little involved and will not be discussed here. The reader is referred to [16] for further details about the H-measure as well as a pointer to R code implementing it.

It is worth noting, however, that Hand [16]'s criticism was recently challenged by Flach et al. [17], who found that the criticism may only hold when the AUC is interpreted as a classification performance measure rather than as a pure ranking measure. Furthermore, they found that under that interpretation, the problem appears only if the thresholds used to construct the ROC curve are assumed to be optimal. The model dependence of the AUC, they claim, comes only from these assumptions. They argue that in fact, Hand [16] did not need to make such assumptions, and they propose a new derivation that, rather than assuming optimal thresholds in the construction of the ROC curve, considers all the points in the dataset as thresholds. Under this assumption, Flach et al. [17] are able to demonstrate that the AUC measure is in fact coherent and that the H-measure is unnecessary, at least from a theoretical point of view.

An interesting additional result of Flach [17] is that the H-measure proposed by Hand [16] is, in fact, a linear transformation of the value that the area under the cost curve would produce. With this simple and intuitive interpretation, the H-measure would be interesting to pit against the AUC in practical experiments, although, given the close relation between the ROC Graph and cost curves, it may be that the two metrics yield very similar if not identical conclusions.

The next section suggests yet another metric that computes an Area under the Curve, but this time, the curve in question is a PR curve.

8.4.5.2 AU PREC and I AU PREC The measures discussed in this section are to PR curves what AUC is to ROC analysis or what the H-measure (or a close cousin of it) is to cost curves. Rather than a single one, two metrics are presented as a result of Landgrebe et al. [14]'s observations that in the case of PR curves, because precision depends on the priors, different curves are obtained for different priors. Landgrebe et al. [14], thus, propose two metrics to summarize the information contained in PR curves: one that computes the Area Under the PR curve for a single prior, the *AU PREC*, and one that computes the Area Under the PR curve for a whole range of priors, the *I AU PREC*. The practical experiments conducted in the paper demonstrate that the AU PREC and I AU PREC metrics are more sensitive to changes in class distribution than the AUC.

8.4.5.3 B-ROC The final experimental method that we present in this chapter claims to be another improvement on ROC analysis. In their work on Bayesian-ROCs (B-ROCs), Cardenas and Baras [18] propose a graphical method that they claim is better than ROC analysis for analyzing classifier performance on heavily imbalanced datasets for three reasons. The first reason is that rather than using the true and false positive rates, they plot the true positive rate and the precision. This, they claim, allows the user to control for a low false positive rate, which is important in applications such as intrusion detection systems, better than ROC analysis. Such a choice, they claim, is also more intuitive than the variables used in ROC analysis. A second reason is similar to the one given by Landgrebe et al. [14] with respect to PR curves: B-ROC allows the plotting of different curves for different class distributions, something that ROC analysis does not allow. The third reason concerns the comparisons of two classifiers. With ROC curves, in order to compare two classifiers, both the class distribution and misclassification costs must be known or estimated. They argue that misclassification costs are often difficult to estimate. B-ROC allows them to bypass the issue of misclassification cost estimation altogether.

8.5 CONCLUSION

The purpose of this chapter was to discuss the issue of classifier evaluation in the case of class-imbalanced datasets. The discussion focused on evaluation metrics or on graphical methods commonly used or more recently proposed and less known. In particular, it looked at the following well-known single-class focus metrics belonging to the threshold category of metrics: sensitivity and specificity, precision and recall, and their combinations: the G-mean and the F-measure, respectively. It then considered a multi-class focus threshold metric that combines the partial accuracies of both classes, the macro-average accuracy. In the threshold metric category, still, other new and more experimental combination metrics were surveyed: mean-class-weighted accuracy, optimized precision, the AGms, and the IBA. The chapter then discussed the following methods and metrics belonging to the ranking category of methods and metrics: ROC analysis, cost curves, PR curves, and AUC. Still in the ranking category, the chapter discussed newer and more experimental methods and metrics: the H-measure, Area under the Precision-Recall Curve and Integrated Area under the ROC Curve, and B-ROC analysis.

Mainly because there has been very little discussion on other aspects of classifier evaluation in the context of class imbalances than evaluation metrics and graphical methods, this chapter did not delve into the issues of error estimation and statistical tests, which are important aspects of the evaluation process as well. We will now say a few words about these issues that should warrant much greater attention in the future.

In the case of error estimation, the main method used in the machine learning community is 10-fold cross-validation. As mentioned in the introduction, it is clear that when such a method is used, because of the random partitioning of the data into 10 subsets, the results may be erroneous because some of the subsets may not contain many or even any instances of the minority class if the dataset is extremely imbalanced or very small. This is well documented, and it is common, in the case of class imbalances in particular, to use stratified 10-fold cross-validation, which ensures that the proportion of positive to negative examples found in the original distribution is respected in all the folds (See [1] for a description of this approach). Going beyond this simple issue, however, both Japkowicz and Shah [1] and Raeder et al. [2] discuss the issue of error estimation in the general case. They both conclude that while 10-fold cross-validation is quite reliable in most cases, more research is necessary in order to establish ideal re-sampling strategies at all times. Raeder et al. [2] specifically mention the class imbalance situation where they suggest that additional experiments are needed to be able to make the kind of statements they made in the general case, when class imbalances are present.

With respect to the issue of statistical testing, we found a single paper—the paper by Keller et al. [19]—that looked at the issue of class imbalances. In this paper, the authors show that a particular test, the bootstrap percentile test for the difference in F_1 measure between two classifiers, is conservative in the absence of a skew, but becomes optimistically biased for close models in the presence of a skew. This points out a potential problem with statistical tests in the case of imbalanced data, but it is too succinct a study for general conclusions to be drawn. Once again, much more research would be necessary on this issue to make stronger statements.

We conclude this chapter by mentioning one aspect of classifier evaluation that was not discussed so far: the issue of evaluation in imbalanced multi-class classification problems. There have been a number of extensions of some of the previously discussed methods for the case of multiple-class-imbalanced datasets. On the ranking method front, these include multi-class ROC graphs, which generate as many ROC curves as there are classes [20]; multi-class AUC, which computes the weighted average of all the AUCs produced by the Multi-class ROC graph just mentioned; and a skew-sensitive version of this Multi-class AUC [21]. On the threshold-metrics front, two metrics have been used in the community: misclassification costs and a multi-class extension of the G-mean (See references to these works in [5] and a description in [3].)

8.6 ACKNOWLEDGMENTS

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